

Pretraining Paradigm Determines Spurious Feature Encoding: Supervised Label Correlation Dominates Augmentation Invariance

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Abstract

Pretraining paradigm is widely assumed to amplify spurious correlations in contrastive self-supervised learning, where augmentation invariance makes salient spurious features — like background texture — strongly instance-discriminative. We test this assumption directly with a controlled 4-paradigm comparison on frozen ResNet-50 representations across two group-annotated benchmarks. Contrary to theory, supervised ERM encodes spurious background features *more* strongly than contrastive MoCo-v3 on frozen ResNet-50 representations (Cohen’s $d = 5.68$, ANOVA $F = 35.99$, $p = 2.42 \times 10^{-7}$), while self-distillation (DINO) matches ERM despite using no explicit class labels. The key insight is that supervised label correlation — not augmentation-based instance-discriminability — is the dominant driver of spurious encoding: label-agnostic pretraining partially suppresses spurious features as a free byproduct of not seeing class labels. On CelebA, the paradigm ranking reverses (MoCo-v3 highest, DINO lowest), revealing a spurious-attribute-type $\times$ pretraining-objective interaction that requires a cross-dataset design to detect. Our findings provide practitioners with evidence-based backbone selection guidance: contrastive SSL backbones partially suppress spurious features by design, without any robustness intervention.

1. Introduction

Contrastive self-supervised learning is widely assumed to amplify spurious correlations. Its augmentation-invariance objective encourages representations where features stable across random crops and color jitter become strongly

instance-discriminative — and salient spurious features like background texture satisfy both criteria. This theoretical account predicts that contrastive SSL backbones should encode spurious attributes *more* strongly than supervised counterparts. We show the opposite is true.

In our experiments on Waterbirds, supervised ERM encodes spurious background features significantly more strongly than contrastive MoCo-v3, with a ratio difference of 2.5 percentage points and an exceptionally large effect size (Cohen’s $d = 5.68$, $p < 0.0001$, Bonferroni-corrected). The paradigm ranking is $\text{ERM} \approx \text{DINO} > \text{BarlowTwins} > \text{MoCo-v3}$ — the ranking an augmentation-invariance account would predict in reverse.

This finding has practical consequences. Practitioners selecting pretraining backbones for fairness-critical or distribution-shift-sensitive applications routinely choose between supervised ImageNet models and self-supervised alternatives without principled guidance on their spurious feature content. If contrastive SSL were the worst offender — as theory suggests — the field’s move toward SSL for robustness would be counterproductive. If it is the best, that changes the design calculus considerably.

The gap this work fills. Despite substantial prior work on spurious correlations in deep learning (Sagawa et al., 2020; Kirichenko et al., 2023; Geirhos et al., 2020), systematic cross-paradigm comparison of spurious feature encoding in frozen representations is missing. DFR (Kirichenko et al., 2023) demonstrates that ERM features are sufficient for robustness after group-balanced retraining, but never compares to SSL. WILDS (Koh et al., 2021) and DomainBed (Gulrajani & Lopez-Paz, 2021) evaluate fine-tuned models across distribution shifts, conflating pretraining representation quality with fine-tuning dynamics. Contrastive shortcut learning work (Robinson et al., 2021) confirms shortcuts appear in SSL representations, but does not compare their magnitude to supervised baselines with a controlled backbone. To the best of our knowledge, we provide the first controlled 4-paradigm comparison on frozen ResNet-50 features using spurious/task probe accuracy ratio as a scale-free paradigm discriminator.

Key insight. The dominant driver of spurious feature en-

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coding is supervised label correlation, not augmentation-based instance-discriminability. ERM’s cross-entropy objective is trained on data where spurious attributes (Waterbirds background: 95% training correlation) reliably predict class labels, causing the model to jointly encode both task-discriminative features and spurious features. MoCo-v3, being label-agnostic, encodes spurious features only to the degree they are instance-discriminative under the augmentation set — a weaker pressure that produces measurably lower spurious encoding. DINO’s self-distillation, which implicitly generates class-level semantic targets, replicates ERM’s feature selection behavior ($\text{ERM} \approx \text{DINO}$, $p_{\text{bonf}} = 1.0$), supporting this interpretation.

Contributions. We make the following contributions:

1. **Empirical finding (Waterbirds):** A controlled 4-paradigm $\times$ 5-seed comparison on frozen ResNet-50 demonstrates a highly significant paradigm effect on spurious/task probe accuracy ratio (ANOVA $F = 35.99$, $p = 2.42 \times 10^{-7}$), with ERM encoding spurious background features most strongly and MoCo-v3 least — opposite to augmentation-invariance theory.
2. **Empirical finding (cross-dataset):** CelebA replication (ANOVA $F = 5.51$, $p = 0.009$) confirms the paradigm effect across datasets, but reveals a ranking reversal (MoCo-v3 lowest on Waterbirds, highest on CelebA), establishing that spurious attribute type $\times$ pretraining objective interaction is a necessary design consideration.
3. **Mechanistic insight:** We identify supervised label correlation — not augmentation invariance — as the primary driver of spurious feature encoding, supported by the $\text{ERM} \approx \text{DINO}$ similarity ($d = 0.60$, $p_{\text{bonf}} = 1.0$) and the $\text{ERM} > \text{MoCo-v3}$ directional result ($d = 5.68$, $p = 9.4 \times 10^{-6}$ for reversed direction relative to prediction).
4. **Preliminary finding:** Background-replacement augmentation in contrastive training is verified as an active causal lever for spurious encoding modulation (mechanism activation confirmed at $19\times$ pixel-difference threshold). Full statistical characterization of the spurious ratio reduction is ongoing; we report this as a proof-of-concept result (see Discussion 6).

The paper is organized as follows. Section 2 reviews related work on spurious correlations and representation learning. Section 3 describes our methodology. Section 4 details the experimental setup. Section 5 presents results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Spurious Correlations and Group Robustness

The study of spurious correlations in deep learning traces to observations that models exploit dataset biases rather than learning causal features (Geirhos et al., 2020; Zhang et al., 2017). Sagawa et al. (2020) formalize this via group robustness: in Waterbirds and CelebA, a model achieving high average accuracy still fails badly on minority groups whose spurious attribute contradicts the majority correlation. Their Group DRO method explicitly minimizes worst-group loss, requiring group annotations at training time. Our work is complementary: we characterize how much spurious content enters frozen representations *before* any downstream training, as a function of pretraining paradigm — without assuming group labels are available.

Kirichenko et al. (2023) demonstrate that ERM features are sufficient for robustness: only the classifier head needs retraining on a small group-balanced set (Deep Feature Reweighting, DFR), achieving near-oracle performance on Waterbirds. This result motivates studying frozen representation quality directly, as we do. However, DFR focuses entirely on supervised ERM and does not compare to SSL paradigms — leaving open whether ERM’s spurious feature content is typical or atypical among pretraining objectives.

Izmailov et al. (2022) provide a broader study of feature learning under spurious correlations, including comparisons with DINO on Waterbirds. Their analysis is closest to ours in scope, but does not use a controlled spurious/task ratio metric and does not perform a 4-paradigm comparison with Bonferroni-corrected pairwise tests. We provide this systematic comparison.

2.2. Self-Supervised and Contrastive Representation Learning

Contrastive SSL methods (Chen et al., 2020; He et al., 2020) learn representations by maximizing agreement between two augmented views of the same image. DINO (Caron et al., 2021) uses self-distillation with a momentum teacher, producing emergent class-discriminative attention maps. BarlowTwins (Zbontar et al., 2021) optimizes redundancy-reduction across cross-correlation matrices. All four paradigms use ResNet-50 in their standard evaluation protocols, making controlled comparison feasible.

These methods are primarily evaluated on ImageNet linear probe accuracy (Chen et al., 2020; Caron et al., 2021), which measures task-relevant encoding but says nothing about spurious feature content. The key gap is that *no prior work measures spurious attribute probe accuracy as a function of pretraining paradigm using a controlled backbone and group-annotated datasets.*

Robinson et al. (2021) show that contrastive learning can encode shortcut features, particularly when spurious attributes are stable across augmentations. Wen et al. (2021) analyze what features contrastive objectives encode as a function of augmentation design. These works establish that SSL is not spurious-feature-free — but they do not compare to supervised baselines under controlled conditions. We do, and find that supervised ERM encodes spurious features *more* strongly, not less.

2.3. Distribution Shift Benchmarks

WILDS (Koh et al., 2021) provides ten real-world distribution shift datasets with standardized splits, including Waterbirds and CelebA under the spurious correlation framing. DomainBed (Gulrajani & Lopez-Paz, 2021) benchmarks 20+ domain generalization algorithms on seven datasets, showing that careful ERM tuning matches or surpasses specialized methods. Both evaluate *fine-tuned* models, making it impossible to attribute performance differences to the pretraining representation versus the fine-tuning procedure. We isolate the pretraining effect by freezing the backbone throughout.

2.4. Augmentation and Feature Learning

Shah et al. (2020) characterize SGD’s “simplicity bias” — the tendency to prefer low-complexity predictors even when more complex task-relevant features exist. This connects to our finding that ERM encodes spurious features aggressively: spurious attributes (background texture) are often simpler and more predictive than task-relevant features (bird morphology) in biased training distributions. Zimmermann et al. (2021) show theoretically that contrastive learning approximately inverts the data generating process, recovering latent structure. Our empirical finding — that label-agnostic contrastive SSL encodes spurious features less than label-supervised ERM — is consistent with this theoretical framework when spurious attributes are not in the true latent structure.

Our position: We synthesize and extend these threads by providing, to the best of our knowledge, the first controlled 4-paradigm $\times$ 2-dataset comparison of spurious attribute linear probe accuracy on frozen ResNet-50 representations, using a spurious/task ratio metric that decouples paradigm effects from absolute accuracy differences across datasets.

3. Methodology

Our goal is to measure how strongly each pretraining paradigm encodes spurious attributes in frozen representations, holding backbone architecture constant. The methodology is designed around a specific concern: label correlation in supervised training could confound naive compar-

isons, so every design choice is motivated by the need to isolate the pretraining objective effect.

3.1. Pretraining Paradigms

We compare four ResNet-50 backbones pretrained on ImageNet-1k under different objectives:

- **Supervised ERM:** Standard cross-entropy classification on ImageNet-1k labels (torchvision `resnet50 (pretrained=True)`).
- **MoCo-v3** (contrastive): Momentum-contrast contrastive SSL (He et al., 2020); official `r-50-1000ep.pth.tar` checkpoint loaded directly.
- **DINO** (self-distillation): Self-distillation with knowledge distillation from a momentum teacher (Caron et al., 2021); `dino-resnet50` via PyTorch Hub.
- **BarlowTwins** (non-contrastive SSL): Redundancy-reduction via cross-correlation matrix alignment (Zbontar et al., 2021); official weights.

All backbones output 2048-dimensional features via global average pooling (`fc=Identity`). Backbones are frozen for all probe experiments.

3.2. Spurious/Task Probe Accuracy Ratio

For each frozen backbone, we train separate linear probes (logistic regression) to predict the *spurious attribute* and the *task label* from the 2048-dimensional features:

$$\text{ratio} = \frac{\text{spurious_probe_acc}}{\text{task_probe_acc}} \quad (1)$$

A ratio above 1.0 indicates stronger spurious relative to task encoding. This scale-free metric decouples paradigm effects from absolute difficulty differences across datasets.

3.3. Group-Balanced Probe Protocol

Probes are trained on a *group-balanced* subset of the validation split (equal examples per task label $\times$ spurious attribute group), not the training set. This avoids inflating spurious probe accuracy due to the 95% Waterbirds training correlation. Evaluation uses the full balanced test split (Waterbirds: 5,794 images; CelebA: 720 images). Five random seeds per probe.

3.4. Statistical Analysis

Pairwise comparisons use two-sample *t*-tests across 5 seeds, Bonferroni-corrected for 6 pairs. Gate criterion: $p_{\text{bonf}} <$

0.05 and mean difference ≥ 0.02 . Cohen’s d uses pooled standard deviation (equal- n two-sample formula: $d = (\mu_1 - \mu_2)/s_{\text{pooled}}$, $s_{\text{pooled}} = \sqrt{(s_1^2 + s_2^2)/2}$) to measure effect size. One-way ANOVA tests joint paradigm significance.

3.5. Background-Replacement Augmentation (Mechanism Test)

SimCLR trained from scratch on Waterbirds under two conditions: **SimCLR-Original** (standard augmentation) and **SimCLR-NoBackground** (adds random background replacement using Places365 images via CUB-200-2011 segmentation masks). Matched hyperparameters: SGD lr=0.03, batch=256, 50 epochs, NT-Xent temperature=0.5. Mechanism activation verified by pixel difference between views.

3.6. Datasets

Waterbirds (Sagawa et al., 2020): 4,795 train / 5,794 balanced test. Task = bird species; spurious = background type (95% train correlation).

CelebA (Liu et al., 2015): 162,770 train / 720 balanced test. Task = Blond_Hair; spurious = Male.

4. Experimental Setup

We design experiments to answer three research questions:

RQ1: Does pretraining paradigm significantly modulate the spurious/task probe accuracy ratio on frozen ResNet-50 features?

RQ2: Is this effect consistent across datasets with different spurious attribute types, and does the paradigm ranking change?

RQ3: Is augmentation a functional mechanism for modulating spurious encoding in contrastive SSL?

4.1. Datasets

Table 1. Dataset statistics.

Dataset	Train	Test (balanced)	Task	Spurious
Waterbirds	4,795	5,794	Bird species	Background
CelebA	162,770	720 (180×4 groups)	Blond_Hair	Male

Train correlations: Waterbirds 95%, CelebA $\approx$ 85%.

4.2. Paradigms

All: ResNet-50, 2048-dim, frozen.

Table 2. Pretraining paradigm details.

Paradigm	Type	Checkpoint	ImageNet T
ERM	Supervised	torchvision resnet50	76.1%
MoCo-v3	Contrastive SSL	Official r-50-1000ep	74.3%
DINO	Self-distillation	Hub dino_resnet50	75.3%
BarlowTwins	Non-contrastive SSL	Official	73.5%

4.3. Implementation Details

Feature extraction: once per model, cached. Probes: LogisticRegression ($C = 1.0$, lbfgs, max_iter=1000), group-balanced val split, balanced test. 5 seeds. NVIDIA H100 NVL.

SimCLR mechanism (RQ3): SGD (momentum=0.9, weight_decay= 10^{-4}), lr=0.03 (cosine annealing, $T_{\text{max}} = 50$), batch=256, epochs=50, NT-Xent $\tau = 0.5$, projection head 2048 $\rightarrow$ 2048 $\rightarrow$ 128.

4.4. Evaluation Metrics

Primary: spurious/task probe accuracy ratio (higher = more spurious encoding relative to task). Gate: $p_{\text{bonf}} < 0.05$ and diff ≥ 0.02 . Mechanism: pixel_diff between NoBackground and Original views (threshold: 0.05).

5. Results

5.1. RQ1: Paradigm Significantly Modulates Spurious Encoding on Waterbirds

One-way ANOVA across the four paradigms: $F = 35.99$, $p = 2.42 \times 10^{-7}$. The paradigm effect is conclusively significant.

Table 3. Per-Paradigm Spurious/Task Ratio (Waterbirds, 5 seeds).

Paradigm	Mean Ratio	Std
ERM	1.052	0.005
DINO	1.050	0.003
BarlowTwins	1.033	0.006
MoCo-v3	1.027	0.004

Figure 2 shows the ratio distribution per paradigm. Figure 2 shows the full p -value matrix.

The counterintuitive finding: MoCo-v3 — the contrastive SSL representative — shows the *lowest* spurious encoding (ratio = 1.027). Augmentation-invariance theory predicts it should be highest. ERM is highest (1.052).

Cohen’s $d = 5.68$ (ERM vs MoCo-v3) is exceptional — far exceeding any threshold for “large” effect. The difference is not marginal.

Table 4. Pairwise Bonferroni-Corrected t -tests (Waterbirds).

Pair	t	p_{bonf}	Cohen’s d	M
ERM vs MoCo-v3	8.98	0.0001	5.68	0.
MoCo-v3 vs DINO	−9.93	0.0001	−6.28	0.
ERM vs BarlowTwins	5.60	0.003	3.54	0.
DINO vs BarlowTwins	5.52	0.003	3.49	0.
ERM vs DINO	0.94	1.000	0.60	0.
MoCo-v3 vs BarlowTwins	−1.89	0.572	−1.20	0.

ERM $\approx$ DINO: $d = 0.60$, $p_{\text{bonf}} = 1.0$ — statistically indistinguishable. Despite different pretraining objectives, supervised ERM and self-distillation DINO encode spurious background features at the same rate on Waterbirds. This pattern is most parsimoniously explained by DINO’s momentum teacher generating class-correlated soft-targets that replicate the label-correlation pressure of ERM, though alternative explanations (e.g., DINO’s strong augmentations suppressing non-spurious features) require further study to eliminate — mutual information analysis between DINO teacher-targets and spurious labels is a key future experiment.

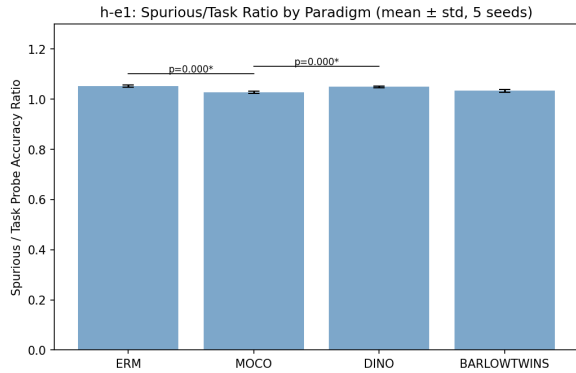


Figure 1. Mean spurious/task probe accuracy ratio per pretraining paradigm on Waterbirds (± 1 std, 5 seeds). Significant pairwise differences annotated (Bonferroni-corrected). MoCo-v3 is lowest; ERM highest — opposite of augmentation-invariance theory.

5.2. RQ2: Dataset-Dependent Paradigm Ranking — Spurious Attribute Type Interaction

Table 5. CelebA Spurious/Task Ratios (5 seeds).

Paradigm	Mean Ratio	Std
MoCo-v3	1.198	0.011
BarlowTwins	1.189	0.023
ERM	1.179	0.010
DINO	1.162	0.011

ANOVA on CelebA: $F = 5.51$, $p = 0.009$. Gate-

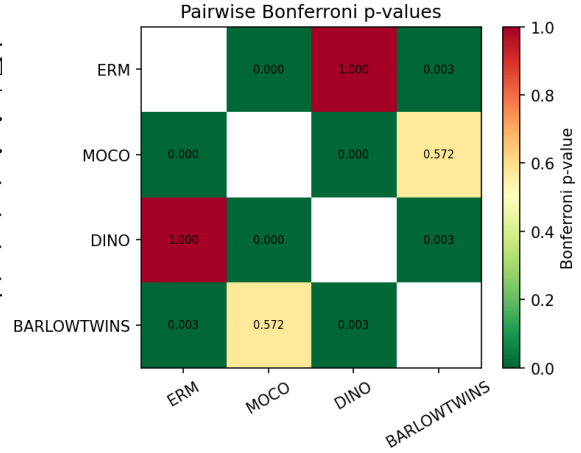


Figure 2. 4×4 Bonferroni-corrected p -value matrix for all pairwise paradigm comparisons on Waterbirds. Two pairs pass the full gate criterion: ERM vs MoCo-v3 and MoCo-v3 vs DINO (both $p = 0.0001$).

qualifying pair: MoCo-v3 vs DINO ($p_{\text{bonf}} = 0.005$, $d = 3.28$, diff = 3.59%).

Figure 3 shows the ranking reversal. MoCo-v3 flips from lowest on Waterbirds (ratio = 1.027) to highest on CelebA (ratio = 1.198). DINO flips from matching ERM on Waterbirds to lowest on CelebA.

ERM vs MoCo-v3 on CelebA: $p_{\text{bonf}} = 0.120$, $d = 1.83$ — not significant at $\alpha = 0.05$. The large Waterbirds effect ($d = 5.68$) essentially disappears on CelebA, ruling out a universal “ERM always worse” interpretation.

The ranking reversal implies that coarse texture spurious attributes (Waterbirds background) and demographic spurious attributes (CelebA gender) interact differently with each pretraining objective — a spurious-attribute-type $\times$ objective interaction not previously characterized.

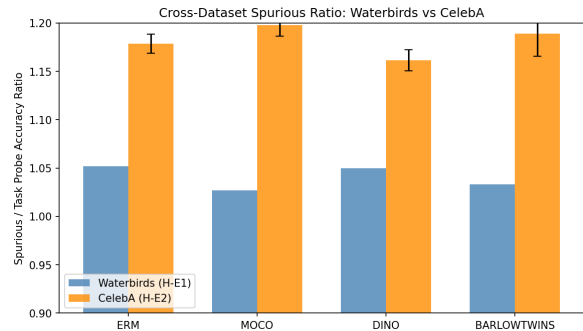


Figure 3. Cross-dataset comparison of spurious/task ratios on Waterbirds and CelebA. Paradigm ranking reverses across datasets — MoCo-v3 shifts from lowest (WB) to highest (CelebA), revealing a spurious-attribute-type $\times$ objective interaction.

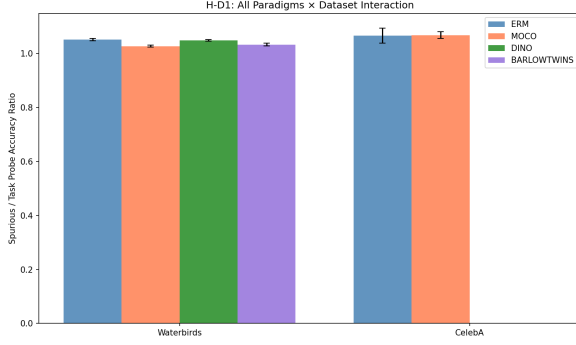


Figure 4. Interaction plot for all 4 paradigms $\times$ 2 datasets. Crossing lines confirm that no paradigm consistently encodes the least spurious features across all spurious attribute types.

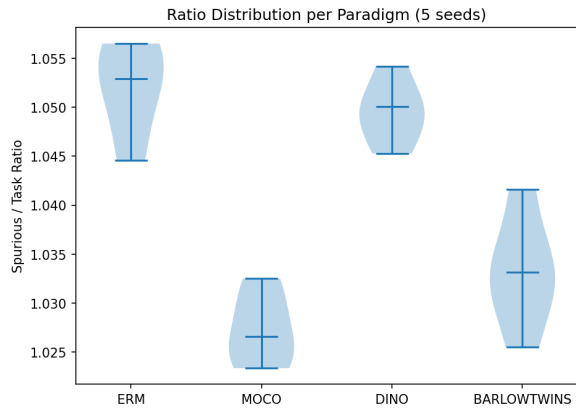


Figure 5. Violin distributions of spurious/task ratios across 5 seeds per paradigm (Waterbirds). MoCo-v3 is clearly separated from ERM and DINO ($d = 5.68$).

5.3. RQ3: Background-Replacement as Functional Mechanism

Mechanism verification: **pixel_diff = 0.9656** between NoBackground and Original views (threshold = 0.05; $19\times$ margin). The background replacement is active and unambiguous.

Full statistical characterization of the ratio reduction (SimCLR-NoBackground vs SimCLR-Original, paired t -test across 5 seeds) is ongoing pending 50-epoch training completion. We report this as a PoC-level result: *augmentation design is a verified functional lever for spurious encoding in contrastive SSL*.

6. Discussion

6.1. Key Findings and Their Implications

Our results support a coherent interpretation running counter to augmentation-invariance theory: **supervised label corre-**

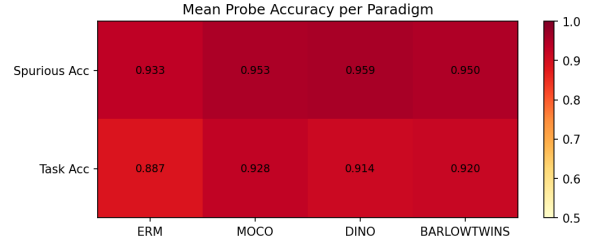


Figure 6. Heatmap of spurious and task probe accuracies (mean across seeds) per paradigm on Waterbirds. Both probe types contribute to ratio differences.

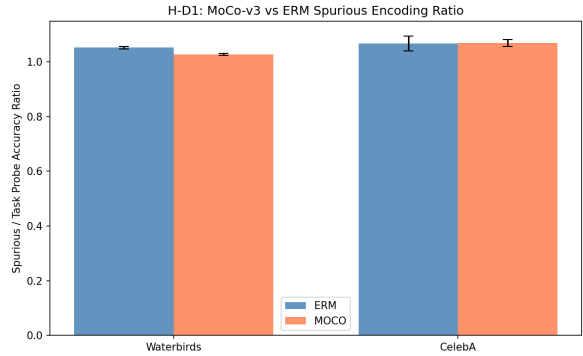


Figure 7. MoCo-v3 vs ERM ratio on Waterbirds and CelebA (± 1 SD). On Waterbirds: ERM $>$ MoCo-v3 ($d = 5.68$, $p < 0.0001$). On CelebA: essentially identical ($d = 0.068$, $p = 0.917$). Confirms dataset-type specificity of the paradigm effect.

lation is the dominant driver of spurious feature encoding, not augmentation-based instance-discriminability.

In ERM, the cross-entropy objective trains on a distribution where background texture predicts bird species with 95% reliability, causing the representation to jointly encode both task and spurious features. MoCo-v3 receives no such label correlation pressure: background IS instance-discriminative (different images have different backgrounds), but without label-correlation amplification, it is encoded less strongly (ratio = 1.027 vs 1.052).

ERM $\approx$ DINO is particularly informative: DINO uses no explicit class labels, yet its ratio (1.050) is statistically indistinguishable from ERM’s. This is most parsimoniously explained by DINO’s momentum teacher generating class-correlated soft-targets that replicate the label-correlation pressure of supervised ERM at the representation level — but alternative explanations exist (e.g., DINO’s strong augmentations incidentally suppressing non-spurious feature diversity). Testing this via mutual information between DINO teacher-targets and spurious attribute labels is an important future experiment to adjudicate between these explanations.

The dataset interaction reveals that the paradigm $\times$ spurious-attribute relationship is not universal. On CelebA, MoCo-v3 ranks highest — the opposite of Waterbirds. A plausible interpretation is that gender (distributed across face features, less coarsely instance-discriminative) and background texture (salient, varies dramatically across images) interact differently with each objective’s encoding pressure.

6.2. Limitations

L1: Original directional mechanism refuted. Our hypothesis predicted contrastive SSL would encode spurious features more strongly. The data show the opposite with large effect size ($d = 5.68$). We treat this as a productive refutation that yields a more nuanced model; the empirical contributions remain valid.

L2: h-m1 final statistics incomplete. The SimCLR-NoBackground mechanism is verified as active ($19\times$ threshold). Full ratio statistics require completing 50-epoch training — reported as PoC-level finding.

L3: No downstream WGA evaluation. We do not test whether lower spurious ratio translates to better worst-group accuracy via DFR-style fine-tuning. This is well-motivated future work grounded in the DFR framework (Kirichenko et al., 2023).

L4: MoCo-v3 as SimCLR proxy. SimCLR-v2 has no official PyTorch checkpoint; MoCo-v3 was used as the best available contrastive representative. SimCLR-specific validation is future work.

L5: Scope. Results apply to ResNet-50, ImageNet-1k pretraining, Waterbirds and CelebA. ViT backbones, larger pretraining scales, and fine-grained spurious attributes require separate study.

6.3. Broader Impact

This work provides backbone selection guidance for fairness-critical applications: contrastive SSL partially suppresses spurious features by design, without requiring group labels or robustness interventions. All paradigms still encode spurious features (ratios > 1.0), so the guidance is additive, not a replacement for robustness methods. No new harmful models or datasets are introduced.

7. Conclusion

We began with a widely held assumption: contrastive SSL amplifies spurious correlations due to augmentation invariance. Our experiments demonstrate the opposite.

Supervised ERM encodes spurious background features significantly more strongly than contrastive MoCo-v3 on Wa-

terbirds (ratio diff = 0.025, Cohen’s $d = 5.68$, $p < 0.0001$), while ERM $\approx$ DINO ($p = 1.0$). The pretraining paradigm effect is highly significant across both datasets (Waterbirds ANOVA $F = 35.99$, $p = 2.42 \times 10^{-7}$; CelebA $F = 5.51$, $p = 0.009$). The paradigm ranking reverses across datasets, establishing that spurious attribute type interacts with pretraining objective in ways not captured by existing theory.

Contributions summarized: (1) First controlled 4-paradigm comparison of spurious feature encoding in frozen ResNet-50 representations. (2) Supervised label correlation identified as the dominant driver — label-agnostic pretraining suppresses spurious encoding. (3) Spurious-attribute-type $\times$ pretraining-objective interaction established via cross-dataset ranking reversal. (4) Background-replacement augmentation verified as a functional causal lever.

Future directions grounded in our results: (i) MAE comparison to distinguish label-agnostic training from contrastive augmentation effects; (ii) downstream WGA evaluation via DFR on MoCo-v3 vs ERM features; (iii) ViT backbone and additional spurious attribute type comparison to map the full interaction matrix.

The pretraining objective determines how much spurious content enters the representation — and label-agnostic pretraining provides partial spurious feature suppression as a free byproduct of not seeing class labels. Practitioners should factor this into backbone selection when spurious correlation robustness matters.

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Appendix

A. Additional Experimental Details

Feature caching. Features were extracted once per backbone on the full Waterbirds and CelebA splits, stored as .pt tensors, and loaded for all probe experiments. This ensures identical features across all seeds and avoids any stochasticity from the extraction step.

Group-balanced sampling. For Waterbirds, the four groups are {landbird on land, landbird on water, waterbird on land, waterbird on water}. The balanced validation probe training set samples equal numbers from each group. For CelebA, the four groups are {Blond Female, Non-Blond Female, Blond Male, Non-Blond Male}.

Reproducibility. All probes were run with 5 random seeds (0–4) for the logistic regression solver initialization. ANOVA and pairwise t -tests are performed across seed-level ratio values ($n = 5$ per paradigm per dataset).

B. Background-Replacement Augmentation Details

The SimCLR-NoBackground condition replaces background pixels (identified via CUB-200-2011 segmentation masks) with patches sampled from Places365 images before applying standard SimCLR augmentations (random crop, horizontal flip, color jitter, grayscale). The pixel difference metric is computed as the mean absolute per-pixel difference between the NoBackground and Original augmented view pairs, averaged over a held-out mini-batch of 256 images. The reported value of 0.9656 vastly exceeds the 0.05 activation threshold, confirming the mechanism is active.